

```
In [1]: import pandas as pd
```

```
df = pd.read_csv('~/.Desktop/189 Project/merged_data.csv')  
df.shape
```

```
Out[1]: (51707, 21)
```

```
In [2]: print(df.columns)
```

```
Index(['realSum', 'room_type', 'room_shared', 'room_private',  
      'person_capacity', 'host_is_superhost', 'multi', 'biz',  
      'cleanliness_rating', 'guest_satisfaction_overall', 'bedrooms', 'dist',  
      'metro_dist', 'attr_index', 'attr_index_norm', 'rest_index',  
      'rest_index_norm', 'lng', 'lat', 'city', 'weekday'],  
      dtype='object')
```

`realSum` : full price for two people and two nights

`room_type` : categorical for private room; entire home/apt; shared room

`room_shared` : dummy for shared rooms

`room_private` : dummy for private rooms

`person_capacity` : maximum number of guests (2-6)

`host_is_superhost` : dummy for superhost status (granted to hosts providing high quality services)

`multi` : dummy if the listing belongs to hosts with 2-4 offers

`biz` : dummy if the listing belongs to hosts with more than 4 offers

`cleanliness_rating` : cleanliness rating (0-10)

`guest_satisfaction_overall` : overall rating of the listing (0-100)

`bedrooms` : number of bed rooms; 0 for studio

`dist` : distance to nearest metro station in kilometers

`attr_index` : attraction index of the listing location

*The `attr_index` for listing j based on K points of interest is calculated as

$$\text{attr index}_j = \sum_{k=1}^K \frac{R_k}{d_{jk}}$$

where R is the number of reviews for attraction k and d_{jk} is the distance between the listing and the point k

(based on data from TripAdvisor; venues without any reviews and outside city borders are excluded; consider only those with more reviews than the first quantile)

`attr_index_norm` : normalized attraction index (0-100)

`rest_index` : restaurant index of the listing location (similar to `attr_index` , but for restaurants)

`rest_index_norm` : normalized restaurant index (0-100)

`lng` : longitude of the listing location

`lat` : latitude of the listing location

`city` : categorical for city; alphabetical order

1: Amsterdam (Netherlands)

2: Athens (Greece)

3: Barcelona (Spain)

4: Berlin (Germany)

5: Budapest (Hungary)

6: Lisbon (Portugal)

7: London (England)

8: Paris (France)

9: Rome (Italy)

10: Vienna (Austria)

`weekday` : dummy for weekday data vs weekend data

In []: